

HITHESH K

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PROFESSIONAL SUMMARY

Electronics and Communication Engineering undergraduate with hands-on experience in Embedded Systems, IoT, AI, Computer Vision, Full-Stack Development and EV Technologies. Skilled in Python, C, React, FastAPI, ESP32 and OpenCV with practical experience developing intelligent automation systems, digital twins and real-time monitoring solutions. Seeking opportunities in Embedded Systems, VLSI, IoT, EV Technology and Product Engineering.

TECHNICAL SKILLS

Languages: Python, C, C++, JavaScript, SQL, Verilog

Frameworks: React.js, FastAPI, WebSockets, Tailwind CSS

Embedded: Arduino, ESP32, Embedded Systems, IoT

Tools: OpenCV, AutoCAD, Git, GitHub, Microsoft Office, Photoshop

Domains: AI, Computer Vision, Automation, VLSI, Digital Twins

INTERNSHIP EXPERIENCE

Hardware Engineering Intern – Smart EV Charging & Battery Twin Dashboard

Jan 2026 – May 2026

- Developed a real-time EV battery monitoring dashboard using React, FastAPI, Python and WebSockets.
- Built telemetry streaming for SoC, SoH, temperature and charging-cycle analytics.
- Implemented interactive visualizations and charging simulations.
- Optimized frontend performance for high-frequency live data updates.

PROJECTS

AI-Controlled Agricultural Rover

- Developed an AI and IoT-enabled autonomous rover for precision agriculture.
- Implemented crop monitoring, disease detection and soil-analysis workflows.
- Integrated ESP32 camera modules, cloud services and embedded hardware.

Drunk Driver Detection System

- Built a smart vehicle safety system using MQ3 alcohol sensors and OpenCV.
- Automated ignition lock mechanism for impaired driving detection.
- Integrated GPS/GSM emergency alert system for real-time notifications.

RELEVANT COURSEWORK

Digital Electronics • Embedded Systems • Microcontrollers • Analog Electronics • Digital Communication • Signals & Systems • VLSI Design Fundamentals • Computer Networks

ENGINEERING COMPETENCIES

- Embedded System Design and Development
- IoT Integration and Sensor Interfacing
- Real-Time Data Processing
- Computer Vision Applications
- Hardware-Software Co-Design
- Technical Documentation and Testing

EDUCATION

B.E. Electronics & Communication Engineering

CGPA: 7.20

Jawaharlal Nehru National College of Engineering, Shivamogga

Visvesvaraya Technological University

Expected Aug 2026

Class XII (Computer Science) – 87%

2021

Class X – 92%

2019

CERTIFICATIONS & ACHIEVEMENTS

- FutureSkills Prime Certification (NASSCOM)
- NITTE Hackathon Participant (2025)
- CMRIT Hackathon Participant (2025)
- Karnataka State-Level Sports Tournament Participant
- Completed Campus Recruitment Training in Aptitude and Professional Skills

ADDITIONAL INFORMATION

Operating Systems: Windows, Linux

Languages: Kannada, English, Hindi

Career Interests: Embedded Systems, VLSI, IoT, EV Technologies, Automation, Product Engineering